

PROVENA Trinity

Technical Specification

AI Image Detection System with Multi-Layer Watermarking

Version:	2.0.0 (Trinity)
Date:	February 2026
Status:	Development Ready
Timeline:	6 months
Budget:	\$200,000

1. Executive Summary

PROVENA Trinity achieves 99% detection accuracy even after heavy transformations by combining latent-space watermarking, pixel-space watermarking, and a 4-layer detection ensemble.

Transform	Current (v1.0)	Trinity (v2.0)
JPEG Q=50	5%	99%
Screenshot	0%	99%
Heavy Edits	N/A	85-90%

2. Four-Layer Architecture

Layer 1: Latent Space (90-95%) - Embedded during generation

Layer 2: Pixel Space (95%) - Adversarially trained

Layer 3: Perceptual Hash (99%) - Fast fuzzy matching

Layer 4: CNN+ViT (85-90%) - AI fingerprint detection

Combined: 99% Detection Rate

3. Implementation Roadmap

Phase	Duration	Deliverable
Latent Watermarking	8 weeks	Tree-Ring integrated (90%+ detection)
Pixel + Detection	8 weeks	Full ensemble (99%+ detection)
Production Deploy	4 weeks	Docker + API + Monitoring
Testing & Launch	4 weeks	Security audit + MNC approval

4. Next Steps

This Week:

1. Create branch: feature/trinity-watermarking
2. Set up directory structure
3. Configure GPU environment
4. Begin Tree-Ring implementation

For Antigravity:

Use `antigravity_prompts.md` with PRD as context

Documentation:

- `PRD_PROVENA_TRINITY.docx`
- `task_breakdown.md`
- `ultimate_watermarking_solution.md`